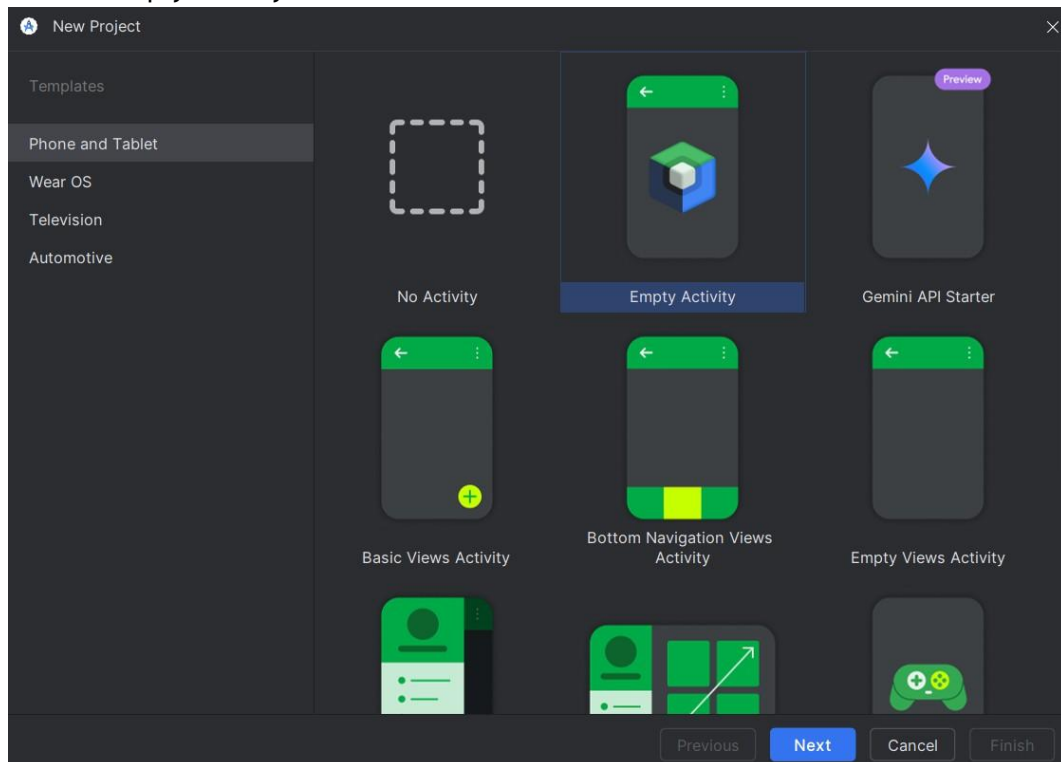


1. Open Android Studio and Create a new Project
2. Click on Empty Activity



3. Write a name for your Application and then click Finish.

In your MainActivity.kt file,

Add/Edit the following code:

```
// PACKAGE DECLARATION - DEFINES THE NAMESPACE FOR THIS APP
```

```
PACKAGE COM.LEARNING.UILEARNAPP
```

```
// IMPORT STATEMENTS - BRINGING IN NECESSARY ANDROID AND COMPOSE LIBRARIES
```

```
IMPORT ANDROID.OS.BUNDLE
```

```
IMPORT ANDROID.WIDGET.TOAST
```

```
IMPORT ANDROIDX.ACTIVITY.COMPONENTACTIVITY
```

```
IMPORT ANDROIDX.ACTIVITY.COMPOSE.SETCONTENT
IMPORT ANDROIDX.ACTIVITY.ENABLEEDGETOEDGE
IMPORT ANDROIDX.COMPOSE.FOUNDATION.IMAGE
IMPORT ANDROIDX.COMPOSE.FOUNDATION.LAYOUT.*
IMPORT ANDROIDX.COMPOSE.MATERIAL3.*
IMPORT ANDROIDX.COMPOSE.RUNTIME.COMPOSABLE
IMPORT ANDROIDX.COMPOSE.UI.ALIGNMENT
IMPORT ANDROIDX.COMPOSE.UI.MODIFIER
IMPORT ANDROIDX.COMPOSE.UI.PLATFORM.LOCALCONTEXT
IMPORT ANDROIDX.COMPOSE.UI.RES.PAINTERRESOURCE
IMPORT ANDROIDX.COMPOSE.UI.TEXT.FONT.FONTWEIGHT
IMPORT ANDROIDX.COMPOSE.UI.TOOLING.PREVIEW.PREVIEW
IMPORT ANDROIDX.COMPOSE.UI.UNIT.DP
IMPORT ANDROIDX.COMPOSE.UI.UNIT.SP
IMPORT COM.LEARNING.UILEARNAPP.UI.THEME.UILEARNAPPTHEME
```

```
// MAINACTIVITY CLASS - THE ENTRY POINT OF THE APP
```

```
CLASS MAINACTIVITY : COMPONENTACTIVITY() {
```

```
    // ONCREATE - CALLED WHEN THE ACTIVITY IS FIRST CREATED
```

```
    OVERRIDE FUN ONCREATE(SAVEDINSTANCESTATE: BUNDLE?) {
```

```
        SUPER.ONCREATE(SAVEDINSTANCESTATE) // CALL PARENT CLASS'S ONCREATE
```

```
        ENABLEEDGETOEDGE() // ENABLE EDGE-TO-EDGE DISPLAY (CONTENT GOES BEHIND
SYSTEM BARS)
```

```
    // SETCONTENT - SETS THE COMPOSE UI FOR THIS ACTIVITY
```

```
    SETCONTENT {
```

```

// APPLY THE APP THEME TO ALL COMPOSABLES INSIDE
UILEARNAPPTHEME {

    // SCAFFOLD - PROVIDES A BASIC MATERIAL DESIGN LAYOUT STRUCTURE
    SCAFFOLD(MODIFIER = MODIFIER.FILLMAXSIZE()) { INNERPADDING ->

        // CALL OUR CUSTOM COMPOSABLE AND PASS THE PADDING FROM
        SCAFFOLD

        SIMPLEAPP(

            MODIFIER = MODIFIER.PADDING(INNERPADDING)

        )

    }

}

}

}

```

```

// SIMPLEAPP - A CUSTOM COMPOSABLE FUNCTION THAT BUILDS THE UI
@COMPOSABLE // ANNOTATION INDICATING THIS IS A COMPOSABLE FUNCTION
FUN SIMPLEAPP(MODIFIER: MODIFIER = MODIFIER) {

    VAL CONTEXT = LOCALCONTEXT.CURRENT

    // COLUMN - ARRANGES CHILDREN VERTICALLY (TOP TO BOTTOM)
    COLUMN(

        MODIFIER = MODIFIER // APPLY THE PASSED MODIFIER

        .FILLMAXSIZE() // MAKE THE COLUMN FILL THE ENTIRE AVAILABLE SPACE

        .PADDING(16.DP), // ADD 16DP PADDING AROUND THE COLUMN

        HORIZONTALALIGNMENT = ALIGNMENT.CENTERHORIZONTALLY, // CENTER
        CHILDREN HORIZONTALLY
    )
}

```

```

        VERTICALARRANGEMENT = ARRANGEMENT.CENTER // CENTER CHILDREN
        VERTICALLY IN THE COLUMN

    ){

        // 1. TEXT COMPOSABLE - DISPLAYS TEXT ON SCREEN

        TEXT(

            TEXT = "WELCOME TO MY APP!", // THE TEXT TO DISPLAY

            FONTSIZE = 24.SP, // TEXT SIZE (24 SCALABLE PIXELS)

            FONTWEIGHT = FONTWEIGHT.BOLD, // MAKE THE TEXT BOLD

            COLOR = MATERIALTHEME.COLORSCHEME.PRIMARY, // USE PRIMARY COLOR
            FROM THEME

            MODIFIER = MODIFIER.PADDING(BOTTOM = 16.DP) // ADD 16DP PADDING
            BELOW THE TEXT

        )

        // 2. IMAGE COMPOSABLE - DISPLAYS AN IMAGE

        IMAGE(

            PAINTER = PAINTERRESOURCE(ID = R.DRAWABLE.MYIMAGE), // LOAD IMAGE
            FROM RESOURCES (DRAWABLE FOLDER)

            CONTENTDESCRIPTION = "APP LOGO", // DESCRIPTION FOR ACCESSIBILITY
            (SCREEN READERS)

            MODIFIER = MODIFIER

                .SIZE(100.DP) // SET WIDTH AND HEIGHT TO 100DP

                .PADDING(BOTTOM = 16.DP) // ADD 16DP PADDING BELOW THE IMAGE

        )

        // 3. TEXTFIELD COMPOSABLE - INPUT FIELD FOR USER TEXT ENTRY

        OUTLINEDTEXTFIELD(

```

```

        VALUE = "ENTER YOUR NAME HERE", // THE TEXT TO DISPLAY (STATIC SINCE
ENABLED=FALSE)

        ONVALUECHANGE = {}, // CALLED WHEN TEXT CHANGES (EMPTY BECAUSE
ENABLED=FALSE)

        LABEL = { TEXT("NAME") }, // LABEL TEXT SHOWN INSIDE THE FIELD

        MODIFIER = MODIFIER

        .FILLMAXWIDTH() // MAKE THE TEXTFIELD FILL THE ENTIRE WIDTH

        .PADDING(BOTTOM = 16.DP), // ADD 16DP PADDING BELOW THE TEXTFIELD

        ENABLED = FALSE // DISABLE EDITING (USER CANNOT TYPE IN IT)

    )

```

// 4. BUTTON WITH ICON - FIRST BUTTON

```

BUTTON(

    // ONCLICK - CALLED WHEN THE BUTTON IS CLICKED/TAPPED

    ONCLICK = {

        // SHOW A TOAST MESSAGE (POPUUP NOTIFICATION)

        TOAST.MAKETEXT(CONTEXT, "BUTTON CLICKED!",
TOAST.LENGTH_SHORT).SHOW()

    },

    // MODIFIER CHAIN FOR STYLING THE BUTTON

    MODIFIER = MODIFIER

    .FILLMAXWIDTH() // MAKE BUTTON FILL THE ENTIRE WIDTH

    .HEIGHT(56.DP) // SET BUTTON HEIGHT TO 56DP (STANDARD MATERIAL
HEIGHT)

    .PADDING(BOTTOM = 12.DP), // ADD 12DP PADDING BELOW THE BUTTON

) {

    // 5. ICON COMPOSABLE - A SMALL ICON INSIDE THE BUTTON

```

```

    ICON(
        PAINTER = PAINTERRESOURCE(ID = R.DRAWABLE.MYIMAGE), // LOAD ICON
        FROM RESOURCES
        CONTENTDESCRIPTION = "ICON", // ACCESSIBILITY DESCRIPTION
        MODIFIER = MODIFIER.SIZE(24.DP) // SET ICON SIZE TO 24DP
    )
    // SPACER - ADDS SPACE BETWEEN THE ICON AND TEXT
    SPACER(MODIFIER = MODIFIER.WIDTH(8.DP)) // 8DP HORIZONTAL SPACING
    // TEXT COMPOSABLE INSIDE THE BUTTON
    TEXT("CLICK ME") // BUTTON LABEL TEXT
}

```

// 6. MODIFIER DEMONSTRATION - SECOND BUTTON WITH DIFFERENT MODIFIERS

```

BUTTON(
    // ONCLICK - CALLED WHEN THE SECOND BUTTON IS CLICKED
    ONCLICK = {
        // SHOW A DIFFERENT TOAST MESSAGE
        TOAST.MAKETEXT(CONTEXT, "ANOTHER BUTTON CLICKED!",
        TOAST.LENGTH_SHORT).SHOW()
    },
    // DIFFERENT MODIFIERS FOR VARIETY AND DEMONSTRATION
    MODIFIER = MODIFIER
        .FILLMAXWIDTH() // FILL THE ENTIRE WIDTH
        .HEIGHT(48.DP) // DIFFERENT HEIGHT (SMALLER THAN FIRST BUTTON)
        .PADDING(HORIZONTAL = 8.DP), // ADD 8DP PADDING ONLY ON LEFT AND
    RIGHT SIDES
    // CUSTOM COLORS FOR THIS BUTTON

```

```

        COLORS = BUTTONDEFAULTS.BUTTONCOLORS(
            CONTAINERCOLOR = MATERIALTHEME.COLORSCHEME.SECONDARY // USE
SECONDARY THEME COLOR
        )
    ){
        // SIMPLE TEXT ONLY (NO ICON IN THIS BUTTON)
        TEXT("ANOTHER BUTTON")
    }
}
}

// GREETINGPREVIEW - PREVIEW FUNCTION FOR ANDROID STUDIO

@PREVIEW(SHOWBACKGROUND = TRUE) // SHOW WITH A BACKGROUND IN THE
PREVIEW

@COMPOSABLE
FUN SIMPLEAPPPREVIEW() {
    // APPLY THEME AND SHOW THE SIMPLEAPP COMPOSABLE
    UILEARNAPPTHEME {
        SIMPLEAPP()
    }
}
}

```